COURSE

Implants

Screws · Plates



Prof Dr Mahmoud Abdel Karim

2 lectures

Watch course

13 November 2026

TRANS PRO

Developed by FAM

Lecturers

Prof Dr Mahmoud Abdel Karim

IMPORTANT

The world is changing very fast, especially with the rapid development of Artificial Intelligence (AI). We believe we should keep up with these changes and use them to make learning easier and more accessible.

This work was created using a computer program developed by us through Vibe Coding, together with several other AI-assisted tools. Using these tools, we were able to create transcripts of lectures, including lectures delivered in Egyptian Arabic, and organize them into a more academic, book-like format.

The aim is to help the listener focus on listening without having to write notes, while having an organized transcript they can easily return to for quick revision.

This transcript is not a replacement for the original lecture or the relevant textbooks and scientific references. The accuracy is generally very good, but AI-assisted transcription can still contain errors. If you are unsure about any point, please check the original lecture or a reliable reference.

This work is completely non-commercial and was created simply to help those seeking knowledge and make educational resources more accessible.

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How to use this book

- Click any line in the Contents to jump to that lecture.
- The time at the bottom right of each page (» 04:57) opens the original lecture on YouTube at that exact moment.
- "↑ Chapter" returns to the lecture's contents; "↑ Book" returns to the book's contents.
- Click any section heading to return to the lecture's contents.
- "(see Fig.)" jumps to the figure; "↑ back to text" under the figure brings you back to where it was cited.
- "Ask AI" beside each heading opens a page that turns that section into a ready question for an AI: simplify it, clarify what the recording left unclear, or check for newer guidelines.
- Orange notes such as [C1] and [Q1] are AI corrections and doubts - mostly where the recording was hard to transcribe. Each lecture lists them after its index of figures.

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Screws: Key Concepts Every Orthopaedic Surgeon Needs to Know

(Screw Anatomy, Biomechanics and Function)

Prof Dr Mahmoud Abdel Karim

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 26) “usually place depth gauge at acute angle” → “depth gauge at obtuse angle is correct”

Introduction and Opening Case Ask AI

Webinar Context Ask AI

- Webinar hosted by **Mina Pharm**
- Host supportive of scientific meetings, projects, interviews
- Speaker: **Mahmoud Abdel Karim** — Lecturer of Orthopaedic Surgery, Cairo
- Topic: basic concepts of screws every orthopaedic trainee needs
- Difficult topic → dedicated webinar on aspects, concepts, principles of **screws**

Opening Case Ask AI

Case

- **74-year-old male**, rheumatoid arthritis
  - Fracture fixed by treating surgeon as shown
  - Plate + screws → **loss of fixation** at 4 months' follow-up
  - No union
- Such cases common → non-unions, failed fixations, various reasons
- Lecture focus:
  - **Why this fixation fails**
  - Screw types; concept of the screw
  - Screw insertion techniques in detail
  - Tips and pitfalls

References Ask AI

- **AO Trauma**; AO Surgery Reference; *AO Principles of Fracture Management*
- AO Trauma e-learning module → **Surgical Screw Fixation** — recommended
- **Orthopaedic Trauma Association**
  - Comprehensive book on elements of fracture fixation
- Based on a lecture by Professor Ahmed Fouad Seif El-Din and Professor Mohamed Gobba

Screw Concept, History and Mechanics Ask AI

Definition Ask AI

- Ship propeller, aircraft propeller, corkscrew → all screws, not only surgical
- Screw = device converting rotational forces → **linear motion**

History Ask AI

- **Archimedes** — father of the screw; invented 3rd century BC
- **Water screw** → lifted water to higher ground
  - Irrigation for Greek city of Alexandria
- **Al-Shadoof** — water pump in Egypt, 1950s (Wikipedia photo)
  - Lower-level channels → higher ground; basis of the screw
- **Carl Hansmann** (German surgeon) → first internal plate fixation
  - Removable steel plate + nickel-plated screws
  - Screws pre-welded to screwdrivers
  - Driver handles attached to plate → protruded through skin
- 1912 — **Sherman**: recommendations on essential properties of bone screws
  - **Sherman plate** → round holes; gold standard until AO design
- 1958 — **Maurice Müller** + Swiss surgeons → AO and internal fixation principles
  - → Evolution of screws and plates

How a Screw Works Ask AI

- Rotational forces → **linear motion**
- Central **core** + surrounding thread
  - Spiral thread → **helical morphology**
  - Rotation → helix slope → advance along **longitudinal axis** of shaft
- Advance → head compressed against bone cortex
  - Head acts as **stop** → resists further longitudinal motion → preload
- Further tightening:
  - Threaded portion advances → minute stretching → **tensile force** in core
  - Balanced by equal **compression force** at head–bone interface
  - Compression → **friction force** — typical of conventional plates
  - Figure: vertical arrow = tensile force along screw, balanced by compression
  - **Friction at bone–implant interface** → resists displacing forces
  - Friction at bone–screw interface → counteracts external deforming forces

Screw Anatomy: Head and Locking Head Ask AI

Parts of a Screw Ask AI

- Parts to list when asked:
  - **Head**, shaft, threads, tip
  - Inner diameter, outer diameter, **thread depth**
  - **Pitch** and lead — two key aspects

Functions Ask AI

Head

- Attachment of rotational screwdriver
- **Torque** application → spread over six points of contact
- Arrest of forward motion (stop) → **head diameter > outside thread diameter**
- Torque → compression → fracture compressed, head pressed on bone → friction
- Prevents screwdriver slippage → avoids head distortion / **stripping**
- Better directional control during insertion
- Good connection → **blind insertion and removal**
  - **MIPO** → fracture unopened; small incision over screw; driver aligned with head

Head Recesses Ask AI

Head

- **Recesses** → secure grip; types:
  - **Slotted** — old designs; regular and carpentry screws
  - **Cruciate (cross-head)**
  - **Phillips** — rare now; some mini plates / mini screws
  - **Hexagonal (Allen)** — pronounced "Allen", not "Alan K"
  - **Torx / Stardrive**
- **Hexagonal recess** — most popular, most commonly used
  - Good lateral guidance; blind insertion and removal
  - Excessive torque → **stripping** → screw head expansion
    - Cause → worn-out screwdriver
  - Check **screwdriver tip** quality regularly before use
  - Worn-out screwdriver → commonest problem in some hospitals → avoid
- **Stardrive** — more advanced; advantages over hexagonal
  - Less prone to stripping → driving force transmitted more directly
  - **Better torque transmission**
  - Minimal stress; least socket deformation
  - Most current plates and screws → **Stardrive recess**

Locking Head Screw Ask AI

LHS

- **Threaded head** = locking head screw
  - Head threads engage **reciprocal threads** in plate hole
  - Advance → **mechanical couple** with plate; no reliance on compression
- Locked into threaded plate hole → **no plate–bone contact** → no compression
- Locks against tilting → **angular stable construct**
  - **Internal fixator principle** — like external fixator, but internal
  - Plate **not pressed against the bone**
- Stability → locking mechanism, not friction
  - Analogous to **external fixator principle**
  - LHS and **Schanz screw** stability → bending stiffness + mechanical couple
- **Combination hole** → locking head screw or conventional screw
- External load → not by friction → **interlocking** interface
  - → **Bending moments at screw head–shaft junction**
- Exam question (Master's, MD, FRCS) → conventional vs locking plate

Feature	Conventional plate	Locking plate
Stability source	Friction: plate underside ↔ bone surface	Mechanical coupling: locking screw ↔ plate
Plate–bone compression	Yes	No
Plate–bone contact	Yes	No
Load transfer	Friction force	Interlocking; bending moments at head–shaft junction
Principle	Friction (compression)	Internal fixator

- Difficult points deliberately repeated
- Questions at end → chat or verbal

Screw Anatomy: Countersink, Shaft and Run-Out Ask AI

Countersink Ask AI

Screw

- Screw handed + "countersink?" → part of the screw, not the instrument
  - **Countersink** instrument exists — different meaning
  - Part directly **under the head** = countersink
- Shapes: conical or hemispherical
  - Usually **hemispherical** → angulation in all directions
    - Tilted in plate hole → maintains **concentric contact** with plate underside
  - **Conical** → must be centred in plate hole
- **Maximum screw–plate angle** in DCP (MCQ):
  - 25° — horizontal plane
  - 7° — transverse plane

Shaft (Shank) Ask AI

- **Shaft (shank)** — below head and countersink
  - Smooth link: head → thread
  - Almost absent in standard **cortical bone screws**
  - Present in **lag screws** and cancellous screws
  - Distinct shank in lag / cancellous screw — common confusion

Run-Out Ask AI

- **Run-out** = transition shaft → thread
  - Abrupt diameter change → **stress riser**
  - Excessive torsional load → screw breaks here
- Breakage causes → **incorrectly centred hole**; hole not perpendicular to plate
- Formerly run-out diameter = **outside diameter**
  - Outside (outer) vs inner (core) diameter → see below
  - Forced into smaller pilot hole → **bone expansion** stresses
  - **Undercutting** (reduced run-out diameter) → eliminated oversizing
- Breakage at run-out → **spiral configuration**
  - → failure under **torsional load**

Thread, Diameters and Thread Depth Ask AI

Thread Ask AI

- Screw = **inclined plane**
  - Helical inclined plane wraps around screw axis
  - Applied effort → force moving screw around its axis
  - → **tension in the screw** → compression on fracture surface
  - Threads → **purchase** in bone
- **Standard orthopaedic screw → single-threaded**
  - Pitch vs lead → frequently confused; see below

Diameters Ask AI

- Two key diameters
- **Core diameter (inner diameter)**
  - Narrowest diameter, across thread base → yellow arrow
  - Determines **pilot hole** drill
  - Lag screw → larger drill = **gliding hole**; smaller drill = pilot hole
- Strength relations (MCQ):
  - **Bending strength** ∝ core diameter³
  - **Tensile strength** ∝ core diameter²
  - Theory → clinical applications
- **Thread diameter** = outside / outer / nominal diameter
  - "4.5", "3.5" = thread diameter → **nominal diameter** (AO)
  - Across maximum thread width
  - = Larger drill in lag screw → **gliding hole**
  - **Larger outside diameter → greater pull-out resistance**
  - Orange arrow (vs yellow = core)

High Yield

- Bending strength ∝ core diameter³
- Tensile strength ∝ core diameter²
- Pull-out strength ∝ outer diameter²
- Core diameter → pilot hole; outer diameter → gliding hole and tap

Thread Depth Ask AI

- **Effective thread depth** = thread projection from core, each side
  - Maximal spiral depth available for **bone purchase**
  - Grips bone → drives screw forward on rotation
- Calculation:
  - Outer − core = thread on both sides (orange − yellow arrows)
  - **Thread depth = (outer diameter − core diameter) / 2**
  - One thread depth per side
- **Cancellous screws** → wider thread depth
  - Cancellous bone → weaker, less dense than cortical
  - → Need higher **pull-out strength**
  - **Deeper thread depth** → more holding, anchorage, bone engagement

Pitch, Lead and Screw Strength Ask AI

Pitch and Lead Ask AI

- Commonly confused terms → definitions and applications
- Engineering graphs used to illustrate
- **Pitch** = distance between adjacent threads
  - Figure → distance between red-arrow lines
  - Larger pitch → larger lead
- **Lead** = distance travelled per complete 360° turn
- Standard orthopaedic screw single-threaded → **pitch = lead**
- **Single start** (one start, one crest) → pitch = lead
- Multi-start → two, three or four starts
  - Two-start → distance = twice the pitch
  - **Screw start** = number of independently running threads on shaft
  - **Double-start screws** → two ridges wrapped around core
- Multi-start → pitch one → lead double or triple
  - **Lead = pitch × number of starts**
- Figure: green = one start; yellow + green = two; + blue = three
  - Some screws → four starts

Exam Pearl

- Pitch = distance between adjacent threads
- Lead = distance travelled per 360° turn
- Single-threaded → pitch = lead
- Lead = pitch × number of starts

Fine versus Coarse Pitch Ask AI

- Pitch → **fine** or coarse
  - Shorter thread distance → finer; longer → coarser
- **Cortical bone screws → fine pitch**; cancellous bone screws → coarse pitch
- Finer pitch → more thread turns per cortex depth → ↑ **pull-out resistance**
- Cortical and **locking screws** → fine pitch → maximal pull-out resistance
  - Not in cancellous screw
- Cancellous compensation → **pull-out strength outer diameter²**

Feature	Fine pitch	Coarse pitch
Thread distance	Small	Large
Screws	Cortical, locking	Cancellous
Pull-out mechanism	More threads engage cortex	Larger outer diameter, deeper thread

Pull-Out Strength in Elderly Bone Ask AI

- Exam: most fracture patients now elderly → how to fix?
- Maximise **pull-out strength**:
  - Large difference outer − inner diameter
    - ↑ Outer, ↓ inner
  - → More **bone under the threads**
- = cancellous screw design → increased thread depth
  - **Fine pitch** → more threads → more cortex engaged
- Denser cortex → **young bone** vs old; cortical vs cancellous
- **Most important factor in pedicle screw pull-out strength**
  - → Bone quality and degree of osteoporosis

Resisting Fatigue Failure Ask AI

- ↑ **Core diameter** → resists fatigue failure
  - Bending strength core diameter^3
  - Minor diameter × 2 → **bending strength × 8**
  - Little or no effect on pull-out strength
- Frequent exam topic

Working Length Ask AI

Screw

- **Working length** = threaded zone anchored in bone
  - Figure → thread anchored in bone cortex
  - Influences stress at **bone–screw interface**
- ↑ with **thick cortical walls** or bicortical purchase
- ↓ with **thin osteoporotic bone** or monocortical screws
- Long preferred → short gives **high interface stress**

Feature	Short working length	Long working length
Causes	Thin cortex (osteoporotic); monocortical screw	Thick cortex; bicortical insertion
Interface stress	High	Low
Verdict	Unfavourable	Preferred

Thread Design Ask AI

- Two profiles:
  - **V-shaped thread** (older) → slanted both sides
  - **Buttress thread** → slanted one side, near right angle other
    - Slanted **leading edge**; trailing edge perpendicular, faces head
- Buttress → **less likely to loosen**; greater long-term pull-out strength (argued)
- **Buttress thread with rounded base** → further improvement

Feature	V-thread	Buttress thread
Profile	Slanted both sides	Slanted leading edge; trailing edge perpendicular
Pull-out (same outer diameter)	Same	Same
Forces at bone–thread junction	Compression + shear	Mainly compression
Corner stress	More — sharp corners	Less — rounded corners
Long term	—	Less loosening; greater pull-out strength

Screw Tips Ask AI

- Tip types:
  - Conventional round **blunt tip**
  - Self-tapping tip
  - Self-tapping self-drilling tip
- Ordinary cancellous screws → **spiral tip**, corkscrew-like

Non-Self-Tapping Tip Ask AI

- Blunt, not sharp
  - After drilling → **tap** ("threader") creates channel
  - **Threads pre-cut with tap, length for length**
- **Greater pull-out resistance** than fluted self-tapping tip
  - Better purchase in distal cortex
  - Conventional screw → only after tapping

Self-Tapping Tip Ask AI

- Thread-cutting device at screw end → **cutting flutes**
  - Bevelled, sharp → cut path for threads
  - Flute channels → transport bone chips out
- AO book → tap even self-tapping screws in **hard cortical bone**
  - Especially young patients
  - Concern for screw breakage → small sizes (reported as 3.5 and 2.0 mm)

Corkscrew, Trocar and Schanz Tips Ask AI

- **Corkscrew tip** → pushes spongy cancellous bone aside, no cutting
  - Compresses trabecular bone → threads its way through
- **Trocar tip** → some older designs; not true thread cutting
- **Schanz screws** — common misuse:
  - Power drill → straight into bone as a drill → wrong
  - **Drill bit first** → not true thread cutting; bone may fracture
  - Exception → self-tapping self-drilling Schanz screws
- **Self-drilling external fixator Schanz screws**
  - Cut drill hole + thread channel
  - Usually **monocortical insertion**
  - Associated screw failure → largely avoided now
- **Engage far cortex without protruding**
  - Protrusion → injury to soft tissues beyond cortex
  - Stop just short of far side

Screw Types Ask AI

- Classification:
  - **Cortical** vs cancellous
  - **Self-tapping** vs non-self-tapping
  - **Fully** vs partially threaded
  - **Cannulated**
  - **Headless**
  - **Locking head**
- Most asked and used → cortical, cancellous (+ locking head)

Cortical versus Cancellous Ask AI

- Cortical bone → **more dense**, small thickness
- Cancellous bone → **more porous**, less dense, larger volume
- Cancellous screw → **thread-forming** → no tapping
  - If ever tapped → **near cortex only**; weak bone may break
- Material → **stainless steel** or titanium
- Thread length → **fully** or partially threaded

Feature	Cortical screw	Cancellous screw
Bone	Cortical: dense, small thickness	Cancellous: porous, less dense, larger volume
Pitch	Smaller	Larger
Thread depth	—	Deeper → more bone engagement
Tapping	Tap hole, or self-tapping screw	Thread-forming → not needed; if ever, near cortex only

Cannulated Screws Ask AI

- Placed **over a guide wire** → wire positioned → drill over → screw
- ↑ **Inner diameter** for same outer → compensates cannulation
  - Ordinary 4.5 screw → core 3.1–3.2
  - Internal channel removes core → core enlarged
  - Bending stiffness preserved → **screw strength minimally affected**
- Cancellous bone → **self-cutting self-tapping** cannulated screws
  - Most companies and new screws → already so

Double-Pitch (Differential-Pitch) Screws Ask AI

- **Herbert screw** — best known
  - Specialised → **interfragmentary compression**
  - Larger pitch → larger lead → travels further in bone
  - Upper ("headless") part → **fine pitch** (upper yellow arrow); lower → other pitch
- Applications:
  - Small bones → **scaphoid** (self-compressing double-pitch screws)
  - Intra-articular fractures → **femoral head fractures**
- Mechanism → **interfragmentary compression by differing thread pitch**
  - Proximal **fine pitch** → travels less far
  - Distal **coarse pitch** → travels further
  - → Fragments drawn together → compression across fracture plane
  - Distal → proximal compression, e.g. **scaphoid**

Locking Screws Ask AI LHS

- Lock into plate
  - Threaded underside of head → **fixed-angle construct**
  - **Larger core diameter** → ↑ bending strength; load over larger bone area
  - **Shallow thread** with blunt edges
  - **Fine pitch**
    - More threads → better strength, interface, **bone engagement**
- **Uniaxial** or polyaxial mechanism
- No reliance on **plate–bone friction** → larger core, smaller thread depth

Variable-Angle Locking Screws Ask AI LHS

- **Angle guide** — metal cone → angulated drilling on entry
- **15° each direction** → 30° total angulation
- Head undersurface:
  - Variable-angle → **hemispherical / spherical**
  - Fixed-angle → **conical**
  - Look similar → **must not be mixed** (exam point)
- **NCB** (another company):
  - Ordinary unthreaded head; threads on **inner surface of hole**
  - Screw angled → 15° each side of cone
  - Locked by **cap** screwed onto threads

Acutrak Screws Ask AI

- **Acutrak screws** → intra-articular injuries, upper or lower limb
  - **Variable thread pitch**; self-tapping
  - Pitch differential → compression at fracture site
  - Very handy

Bioabsorbable Screws Ask AI

- **Bioabsorbable screws** → alternative to stainless steel / titanium
  - Early famous use → **anterior cruciate ligament reconstruction**
  - Overcome metal drawbacks → e.g. **artefacts on MRI**
  - Composition → **alpha-polyesters** / poly-alpha-hydroxy acids

Functions of Screws Ask AI

- Exam → list functions; never open with a number
  - Plate, lag, position, locking head, and more

Plate, Lag, Position and Locking Functions Ask AI

- **Plate screw** → compresses implant onto bone surface (friction)
- **Lag screw** → interfragmentary compression (detailed below)
- **Position screw**
  - **Holds two bones in correct relationship**
  - No lagging or compression → **thread hole only, no gliding hole**
  - Commonest use → **syndesmotic screws**
  - Articular fracture with gap / defect (e.g. distal humerus) → no compression possible
- **Locking head screw** → **angular stable construct**
- **Interlocking screws** → IM nail, tibia / femur, proximal + distal
  - ↑ Stability; maintain **length, alignment and rotation**
- **Anchorage screw**
  - **Screw head → anchor for wire loop** (tension band) or strong suture
  - Image → anchor screw for cerclage

Push–Pull and Reduction Screws Ask AI

- **Push–pull screw** — look out for it in surgery
  - With **articulated tension device** (plates lecture, QR)
  - **Temporary point of fixation**: screw short distance from plate
    - Device hook in last plate hole; other side fixed by screw
  - Turning device screw → **compression** (possibly distraction)
- Device placement (fellowship / FRCS exam):
  - **Acute angle** → fracture closes, no shear
  - Obtuse angle → **shear force**, failed compression, overlap (over-compression)
- **Reduction screw**
  - Conventional screw through plate → pulls fragment → **indirect reduction**
  - Also open procedures; **MIPO** with locking compression plate

Poller (Blocking) Screws Ask AI

Poller

- **Poller screw** — bollard concept; merits own lecture
  - **Bollards** → street barriers rising from ground (malls, compounds, hotels)
  - Regulate traffic → force a track; Europe → **retractable bollards**
  - **Krettek**, 1999 → described / popularised ; screws guide nail like bollards
- Function → **block main fragment movement around IM nail**
  - Indication → **wide metaphyseal canal** — proximal / distal, femoral / tibial
  - Narrows canal → nail forced along desired trajectory
  - → Reduction + **central nail position**
- Technique (more in IM nail lecture):
  - Lines → metaphyseal fragment axis + fracture line
  - **Proximal** line → tibia; distal line → femur
  - Proximal tibia → proximal fragment axis + fracture line → acute + obtuse angles
  - **Screw in acute angle** — always
  - Distal femur → distal segment axis + fracture line → **acute angle side**
  - **Nail deflects on screw → reduction** + compression
    - Nail trajectory adjusted by screw

Screw Sizes and Length Ask AI

Size Table Ask AI

Sizes

- **Mini-fragment**, small-fragment, large-fragment screws
- Size table → **pass or fail**; memorise
- **4.5 cortical**: named for outside (thread) diameter; head 8 mm; core 3.1 mm
  - **Pilot hole** drill 3.2 mm; gliding hole 4.5 mm; tap 4.5
- Pilot drill → not 3.5 → **3.2** (core 3.1)
  - **Pilot hole ← inner (core) diameter**
- **3.5 cortical** → tap 3.5 cortical; gliding 3.5 (lag); pilot 2.5
- Examiner trick → 2.7 for pilot? → wrong → **2.5 must be used**
  - 2.7 = outside diameter of **2.7 screw**
- **2.7 screw** → gliding 2.7; pilot 2.0; tap 2.7
- **6.5 cancellous** — standard large cancellous screw
  - Same core as 4.5 cortical → 3.1
  - Unthreaded shaft 4.5; outside diameter 6.5 mm
  - Drills as 4.5 screw → pilot 3.2, gliding 4.5; **tap** 6.5
- Cancellous tapping rarely → **near cortex only**; fragile bone
- **Small cancellous screw** → replacing former 4 mm cancellous screws
  - Core 1.9 → 2 mm drill for pilot hole
  - Unthreaded shaft 2.4 mm
  - Outside diameter 3.5 mm → determines size
- Some cancellous screws → thread lengths 16 and 32 mm ("serration")
- **Small cancellous screws** → fully and partially threaded
  - Partially threaded → thread length overall screw length

Screw	Outside (thread) Ø	Core Ø	Pilot drill	Gliding drill	Tap	Other
4.5 cortical	4.5	3.1	3.2	4.5	4.5	Head 8 mm
3.5 cortical	3.5	—	2.5	3.5	3.5	—
2.7	2.7	—	2.0	2.7	2.7	—
6.5 cancellous	6.5	3.1	3.2	4.5	6.5	Unthreaded shaft 4.5
Small cancellous (replacing 4 mm)	3.5	1.9	2.0	—	—	Unthreaded shaft 2.4

Pitfall

- 4.5 screw pilot hole → 3.2, not 3.5
- 3.5 screw pilot hole → 2.5, not 2.7
- 6.5 cancellous → same drills as 4.5 screw; only tap differs (6.5)

Screw Length Ask AI

Sizes

- **Length = top of head → screw tip**
- Excluding head → wrong
  - **Nominal length** = head top → tip — pass or fail
- Length choice:
  - Too short → incomplete purchase
  - Too long → **soft tissue irritation**
  - Ideal → **just protruding beyond cortex**
- In doubt, no structure at risk → "better long and wobbly than short and loose"
  - Best still → just protruding beyond cortex
  - Standard cortical screws → just beyond **far (trans) cortex**
- **Self-cutting / self-tapping screws** → slightly longer, never shorter
  - Tip flute embedded → bone ingrowth on healing → **removal very difficult**
  - → Project about 2 mm; protect nearby structures

Screw Insertion and Lag Screw Technique Ask AI

Insertion Steps Ask AI

Insertion

- Sequence:
  - **Pilot hole** → core (inner) diameter
  - **Countersink** → direct bone fixation only; not through plate
  - **Tap** → outside diameter; e.g. 4.5 screw → pilot 3.2, tap 4.5
- **Tap = outside (thread) diameter**
- Tap function (frequently asked):
  - Screw = **male helix** ; tap → **female helix**
  - Exact reciprocal → pitch + outside thread diameter
  - Specific tap size = outside diameter
- **Self-tapping screws** → flutes (one, two or more)
  - Cut channel; clear chips → **avoid jamming of bone debris in helix**

Lag Screw Principle Ask AI

Lag

- Better name → **interfragmentary screw**
  - **Compressive force across fracture plane**
- Pilot hole only, no **gliding hole**:
  - Head seats → no compression at fracture level
  - → **distraction across fracture**; won't close
  - **Polarised light model** → stresses across fracture and under head → bone damage
- **Gliding hole first** → outside diameter drill → near cortex
- Undersized gliding hole → same problem
  - 3.5 screw → needs 3.5 drill; set often has 3.2 → wrong
  - **Gliding hole = outside (thread) diameter**
  - Undersized → no compression; head poorly seated; **near cortex stress**
- Lag screw **perpendicular to fracture plane** → otherwise displacement
  - Non-perpendicular → ↓ **compression** or further displacement

Drilling the Gliding and Pilot Holes Ask AI

Lag

- **Anatomical reduction** — must
  - Clamps → **Weber clamp** or other reduction clamp (Verbrugge, as heard)
  - **Hold fragments together** with temporary reduction tool
- **Gliding hole** — larger drill bit
  - = outside diameter → 4.5 screw → **4.5 drill bit**
  - **Near cortex** only, not far cortex; drilled outside in
  - **Inside-out drilling** → rare (books, AO module)
    - Outside-in difficult or fragment small
    - Preserve **biology** → no excessive fragment rotation or stripping
    - Mostly outside in
- **Pilot (thread) hole**
  - **Far (trans) cortex**; centred via **drill guide**
  - 4.5 screw → pilot 3.2 → **3.2 drill guide**
  - Special guide (some hospitals) → extension each end = outside diameter
  - 3.2 sleeve inside 4.5 track → holes **coaxial**, perpendicular to fracture
- Pilot drill reminder:
  - 4.5 screw → 3.2
  - 3.5 → 2.5
  - 2.7 → 2.0
  - Others → **mini screws**

Pilot Hole First Ask AI

Lag

- Usually gliding first; pilot first → **very rare**
  - Indication → **narrow spike at far fragment tip**; coaxial gliding hole hard
  - Needs **special curved drill guide**
    - Curve wrapped to far cortex; 4.5 sleeve on top after pilot → gliding hole
  - Advantage → both holes exactly **in line**
  - Difficult → great skill, experience, special guide
- Why gliding first?
  - Small drill first → large drill may stray
  - **Gliding and pilot holes exactly coaxial**

Countersinking Ask AI

Lag

- **Near cortex** only
- Functions:
  - **Platform for head**, matching head diameter
  - **Spreads head load over cortex** — the key point
  - Removes extra bone → head seats without extra stress
- **Larger head contact area → lower stress**
- **No proper countersinking** → eccentric loading
  - High head–bone stress → microfractures, loosening, less compression
  - Care in **diaphyseal cortex**
- **Too much countersinking** → head into medullary canal
  - No resistance to head → no compression; head sinks
  - **Not all cortical bone removed**
- **Cancellous bone** → thin; countersinking usually not needed
  - Weak → **washer instead of countersink**

Measuring Length Ask AI

Lag

- **Depth gauge** → often difficult in oblique fractures
  - Tip fully out and **engaged completely** → correct measurement
- Placement:
  - **Obtuse angle** → correct *[C1 · transcript: "usually place depth gauge at acute angle"]*
  - Acute angle → **too short**
- Acute angle only possible → **add 2–4 mm**
  - "Inaccurate gauge" blamed → often **inappropriate angle**

Exam Pearl

- Articulated tension device → acute angle side; no shear
- Device on obtuse side → shear, failed compression, over-compression
- Poller screw → acute angle side
- Depth gauge → obtuse angle side; acute → too short

Tapping Ask AI

Lag

- **Tap** → cuts thread channel = outside (major) diameter
  - ↑ Torque fraction converted → **compressive force** across fracture
  - **Precision manoeuvre** → no hand wobble
- **Tap sleeve** — mandatory
  - Hand **toggling** → second helix, another track
- Through gliding hole without turning → **avoids false route of tap**
- Controlled on insertion and withdrawal → **protects soft tissues**
  - **Tap cuts thread on withdrawal as well as insertion**
  - → thread damage; **compromised screw hole**
- **Osteoporotic bone** → usually no tap; if ever, near cortex only

Screw Insertion Ask AI

Lag

- Appropriate-length screw → insert, tighten
  - Tightening on **near cortex** → strong stress; cortex may break
- Through gliding hole → **push, don't turn**
  - Avoids **false start** → wide hole permits wrong direction

Lag by Technique versus Lag by Design Ask AI

Lag

- "Lag" → technique and screw type both:
  - **Lag by technique** → gliding + pilot holes → fracture compression
  - **Lag by design** → lag (shaft) screw
- Proximal tibia → **cancellous screw**
  - Drill **near cortex** only
  - **No countersink** → avoid weakening proximal tibial cortex
  - Steps:
    - Drill → no need to exit far side
    - Depth gauge
    - Tap → near cortex only, if ever
    - Screw
  - Weak cancellous bone → **washer**
- Screw number ← **fracture length**
  - **Distal fibula**, long fracture → 2–3 interfragmentary screws

Drilling and Heat Necrosis Ask AI

Energy and Heat Ask AI

Drilling

- **Drilling → bone deformation**
- **Two-thirds of energy** → cutting; disrupts molecular and atomic bonds
  - Remainder → heat, vibration, sound
- Heat effects:
  - Enzyme deactivation → **alkaline phosphatase**
  - **Collagen** and hydroxyapatite bond degradation
  - **Permanent alteration** in mechanical properties
  - → **Thermal necrosis** (osteocytes)
- Effect on fixation:
  - **Conical damage zone** around drill bit (red cone)
  - Bone death → **loosening of screw anchorage**
- **Dead bone → active culture site for infection**
  - Infection → plate failure; **heat necrosis** one cause

Factors Affecting Heat Necrosis Ask AI

Drilling

- Exam → factors increasing vs decreasing heat necrosis
- Unused drill bit → back to nurse → cleaned during screw insertion
- Key → **consistency**, doing the right thing

Increase heat necrosis	Decrease heat necrosis
Dull drill bit	New, sharp drill bit
Long drilling time	Continuous saline irrigation
Thick bone → excessive thrust or speed	Staged drilling for large holes
Wobbling bit → widened hole	Frequent bit cleaning: withdraw, clean, reinsert

Sharp versus Blunt Drill Bit Ask AI

Drilling

- Exam → identify sharp vs blunt without touching
  - **AO skill lab** trick → look at tip
  - **No light reflection** → sharp
  - **Light reflected on tip** → blunt, overused
- Recommendation → **AO membership**, website, courses

> **Clinical Pearl**
> • Sharp drill bit → no light reflection at tip
> • Blunt drill bit → light reflected on tip

Good Drilling Practice Ask AI

Drilling

- **Straight, sharp drill bit of recommended size**
  - Not 3.2 for 3.5, or vice versa
- **Clean tip and flutes** between drillings
- **Start slowly**; establish and maintain direction
  - Rushing → **thermal necrosis** → complications
- **Drill sleeve** — often skipped
- **Power drill**
- **Simultaneous saline irrigation**
- **Proper drilling depth** → avoid over-penetration
  - More with **blunt drill bit** → harder pushing → more injury

Drill Bit Failure Ask AI

Drilling

- **Avoidable**:
  - Hitting implant → periprosthetic stem or **nail**
  - Excessive **speed and thrust**
- **Unavoidable** → hitting far cortex at unfavourable angle
  - **Pelvis**, long trajectories → gradual manoeuvring; unintended bending
  - → **Rotational bending** → breakage
  - Best example → **column screw of the pelvis**

Plate Span Ratio and Plate Screw Density Ask AI

- Commonly missed in exams → how many screws; how long a plate
- **Plate span ratio** — Gautier and Sommer
  - Plate length = 2–3 × fracture length; preferably 3×
  - 10 cm fracture → 30 cm plate
- **Plate screw density**
  - Long plate → don't fill every hole
  - **Screws / screw holes < 0.5** — roughly half
- Example → **tibia**, 14-hole plate → density 0.43
  - Proximal segment → **every other screw**; 3 of 6 = 50%
  - Fracture zone → **no screws** (relative stability)
  - Distal segment → anatomy forced 3 of 4 = 75%
  - Overall 0.43 < 0.5 → acceptable
- Filling all holes → ignores **relative vs absolute stability**

High Yield

- Plate span ratio → plate 2–3 × fracture length, preferably 3×
- Plate screw density → screws / holes < 0.5
- Relative stability → no screws around fracture zone

Opening Case Review and Summary Ask AI

Why the Plate Failed Ask AI

Case

- Inappropriate **screw distribution**
  - Should be **relative stability** → rheumatoid osteoporotic bone
- Osteoporotic bone → no **absolute stability** with lag screws → doesn't hold
- Unnecessary screws filled proximally
- Most screws **not gripping far side** → more screws needed in that bone
- Poor **alignment**
- **Interfragmentary screw** in comminuted osteoporotic segment → unsuitable
- → **Non-union** → early failure, plate breakage

Revision Fixation Ask AI

Case

- **Double plating**
- ↑ **Plate span ratio** → 3× fracture or more
- **Medial plate** → distal femur medial comminution → second fixation plane
- **Every other screw** where possible
- **No screws in intercalary segment** → relative stability

Summary Ask AI

- **Plates lecture** (3–4 years ago; QR; YouTube) → completes this lesson
- Covered:
  - **Functions** of screws
  - **Screw anatomy**
  - **Types** of screws
  - **Screw insertion technique** → pass or fail

AO Trauma Courses in Egypt Ask AI

- **Four AO Trauma courses** this year
  - **Basic Principles of Fracture Management** → 25–28 June; repeated 1–4 July
  - **Upper Extremity One** (shoulder to elbow) → 27–29; cadavers
  - **Upper Extremity Two** (elbow to wrist) → 30 to 2 July
- Not yet done an AO course → encouraged to attend

Plates Designs and Functions

(Plate Functions, Stability and Fixation Principles)

Prof Dr Mahmoud Abdel Karim

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

- **C1** (p. 41) “introduced in 1969 by Dennis” → “introduced in 1969 by Perren et al.”
- **C2** (p. 52) “other side: acute angle or obtuse angle” → “other side: the obtuse angle”
- **C3** (p. 53) “plate on compression convex side” → “plate on tension (convex) side”
- **C4** (p. 58) “final screw density 0.43%” → “final screw density 0.43 (43%)”
- **C5** (p. 58) “screw density less than 0.5%” → “screw density less than 0.5 (50%)”

Introduction and Overview Ask AI

Introductory Case Ask AI

- Topic → **plates**: designs and functions
- 74-year-old male with **rheumatoid arthritis**
  - Comminuted fracture → junction of distal femoral metaphysis and shaft
  - Butterfly fragment
  - Osteoporotic bone
- Treated with a **distal femoral locked plate**
  - Immediate post-op X-ray → different screw types, orientations, types of stability
  - Follow-up ≈ two months
- Three months → **full weight bearing** allowed → plate failed
  - Why so early? → answered at lecture end

Learning Outcomes Ask AI

- Different **functions of plates**
- Absolute versus relative stability
- Different **forms and designs** of plates
- Tips and tricks → fixation principles using plates
- Scope → only what is relevant to plates

Sources Ask AI

- **AO Trauma** education; teachers at Kasr Al Ainy, England, AO courses, senior AO colleagues
- **AO Surgery Reference** → a must; AO membership advised
- **AO Principles of Fracture Management**, third edition → main basis; must-have for juniors and seniors
- **Orthopaedic Trauma Association (OTA)** website → resident lectures; some material used

The Six Plate Functions Ask AI

- Plate functions:
  - **Neutralization** (protection)
  - Compression
  - Tension band
  - Buttress
  - Bridging (internal fixator)
- Classic teaching → five functions
- AO book third edition → added **reduction** → six functions

Exam Pearl

- Six functions of the plate → very common written and oral question
- Neutralization, compression, tension band, buttress, bridging, reduction

The Surgeon Determines the Function Ask AI

- Question → does each plate (e.g. **DCP** vs LCP) have a fixed function? → no
- Same plate → applied differently → different function
- Function and application → decided by the surgeon, not the plate classification
- Quotation from the AO books

Must-Know

- "The surgeon, not the designer of the plate, determines how a plate will function and how it will be applied."
- Most important take-home message of the lecture

Absolute and Relative Stability Ask AI

AO Principles of Internal Fixation Ask AI

- **Anatomical reduction**
- Stable internal fixation
- Preservation of blood supply
- Ultimate goal → **early, active, pain-free mobilization**

Stability versus Rigidity Ask AI

- **Stability** → fracture withstands functional (physiological) loads
  - Defined by **degree of lack of motion between the fracture fragments**
- Stability → refers to the fracture
- **Rigidity** → implant itself withstands deformation
- Stability = fracture status; rigidity = implant property

Term	Refers to	Definition
Stability	Fracture	Ability to withstand functional (physiological) loads; degree of lack of motion between fragments
Rigidity	Implant	Ability of the implant to withstand deformation

Definitions of Absolute and Relative Stability Ask AI

- **Absolute stability** → no micromotion whatsoever at the fracture site under normal physiological loads
  - Total abolition of micromotion
- **Relative stability** → some motion at the fracture site
  - Sufficient to **stimulate tissue differentiation**
  - Below the critical strain level
- Both terms → familiar from AO courses

Perren Strain Theory Ask AI

- Must be known for the orals
- Motion at fracture site → deformation → **strain** in granulation tissue
- Small gap fracture → strain → **cell disruption**
- Large gap fracture (complex, multifragmentary) → strain distributed among cells
  - Not concentrated on a single cell → each cell experiences less traction force
- Example 1 → simple fracture, anatomical reduction, absolute stability
- Example 2 → simple fracture, small gap, fixed with a **bridging plate**
- Example 3 setting → simple fracture believed anatomically reduced, but not
- Example 4 → **multifragmentary (complex)** fracture, large gap, bridging plate

Fracture and fixation	Gap	Stability	Strain	Healing
Simple, anatomical reduction	None	Absolute	No micromotion	Direct (primary); no external callus
Simple, bridging plate	Small	Relative	High → cell disruption	Delayed or absent → plate failure
Simple, imperfect "anatomical" reduction	Small	Relative	High	Delayed union, non-union, plate failure
Multifragmentary, bridging plate	Large	Relative	Low despite movement	Indirect (secondary); callus formation

Pitfall

- Simple fracture "anatomically reduced" but gap remains → small gap under motion
- Result → **delayed union**, non-union, or plate failure

Absolute Stability Ask AI

Absolute

- Effective abolition of fracture-site movement
  - Fracture surfaces do not displace under functional or physiological loads
- Healing → **primary bone healing** via cutting cones (osteons) crossing the fracture
- Reduced strain → no external callus
- Contrast → relative stability: large gap, movement below critical strain → **secondary bone healing**
- Requirements:
  - **Anatomical reduction using compression** across the fracture site
    - Classic → interfragmentary compression with the **lag screw**, particularly intra-articular
  - **Soft-tissue handling** → extremely important, absolute or relative
  - Preservation of blood supply → critical to success
- Interfragmentary compression methods → beyond scope (screws lecture)
- **Lag screws** → best means of interfragmentary compression; a must in intra-articular fractures
- Compression achievable with a plate:
  - Axial compression
  - Lag screws plus a neutralization plate
  - Dynamic compression
  - **Tension band** plate
  - Buttress plate
- Total abolition of motion → direct healing via cutting cones
- Indications → **intra-articular fractures**, simple metaphyseal fractures (type A)
- Contraindication → **multifragmentary metaphyseal–diaphyseal** fractures
  - Not only contraindicated → cannot be achieved
  - Gathering every fragment → **soft tissue disrupted** → blood supply not preserved

Exam Pearl

- Compression with a plate, not only lag screws:
- Axial compression; lag screws + neutralization plate; dynamic compression
- Tension band plate; buttress plate

Relative Stability Ask AI

Relative

- Full discussion with **intramedullary** methods → beyond scope; applied here to bridging
- Some motion → **initial resorption** at fracture site
- → secondary bone healing (callus formation)
- Not "no reduction" → reduction not anatomical, as anatomical is unachievable
- Requirements → correct restoration of **length, axis, and rotation**
- Multifragmentary comminution → refers to the **metaphyseal part**
  - Intra-articular region → still anatomical reduction with lag screws
  - Metaphysis → no anatomical reduction → relative stability
- **Preservation of blood supply** → still critical in both types
- "Some motion" = **controlled movement**
  - Sufficient to stimulate tissue differentiation → later callus
  - Below critical strain level = below **tolerance limit of the healing bone tissue**
  - Excess motion → disruption → **non-union**
- Classic indication → **multifragmentary or complex fractures**

Feature	Absolute stability	Relative stability
Fracture-site motion	None under physiological loads	Some, controlled, below critical strain
Reduction	Anatomical, with compression	Length, axis, rotation
Healing	Primary, cutting cones	Secondary, after initial resorption
Callus	No external callus	Callus formation
Blood supply	Must be preserved	Must be preserved
Indications	Intra-articular; simple metaphyseal (type A)	Multifragmentary or complex fractures
Not for	Multifragmentary metaphyseal–diaphyseal	—

Example: Complex Humeral Fracture Ask AI

- Multifragmentary (complex) **humerus** fracture → anatomical reduction may be unachievable
- Even if achieved → at expense of soft-tissue disruption
- Complex fractures → always request **CT scan** → hidden comminution on some views
- Fixation option → relative stability concept
- Images → AO Surgery Reference and AO book

Plate Forms and Designs Ask AI

Plate Types and Evolution Ask AI

- Section aimed at juniors
- **DCP** (dynamic compression plate) → 4.5 and 3.5
- 4.5 DCP → narrow and broad
- Practical → at first fixation case, senior resident shows juniors the plates
- 4.5 DCP → femur or tibia; small DCP (3.5) → forearm fractures
- LC-DCP → **low-contact** dynamic compression plate
- Evolution of the plate:
  - DCP
  - → LC-DCP (also 3.5 and 4.5)
  - → **LCP** (locking compression plate)
- Abbreviations → needed for hospital ordering and fellowships abroad
- Reconstruction plates → pelvic and acetabular fractures; 3.5 and 4.5
- Semi-tubular and one-third tubular plates
- Less Invasive Stabilization System (LISS) → discussed later

DCP Ask AI

DCP

- **The name of the plate does not always correlate with its function**
  - Same plate → compression, neutralization, or tension band
- Small DCP (3.5) → classically forearm fractures
- First introduced in 1969 (attributed to Perren et al.) *[C1 · transcript: "introduced in 1969 by Dennis"]*
  - Revolutionary concept of **compression plating**
  - New hole designed for axial compression
- Broad 4.5 → classically femur
- Narrow 4.5 → sometimes humerus
  - Small humerus → cannot accommodate a broad DCP
  - Broad 4.5 → also very often humerus
  - Tibia → 4.5
- Small DCP (3.5) → classic example forearm fractures
- Many plates before the DCP → history on AO website
- Unstable fixation → **fatigue and failure**
  - → strict adherence to the principles of compression
- **Intimate contact with the underlying cortex** → blood supply compromise → fatigue or failure
  - Undersurface straight → no notches or undercutting
- Plate directly on bone → **periosteal blood supply** compromised
  - → plate failure
  - → refracture after plate removal

LC-DCP Ask AI

LC-DCP

- Design change → overcome problems of DCP lying directly on bone
  - Reduced footprint → **notching / undercutting** of undersurface
  - Minimized kinking at screw holes → more contourable
  - Reduced plate fatigue at holes → where fatigue failure used to occur
  - More **screw inclination** → longitudinal and transverse planes
- DCP → **plate hole area less stiff** than area between holes
- → DCP bends only at screw-hole areas
- LC-DCP → **even stiffness** across whole length
  - → lower risk of kinking at screw holes on contouring
- Materials → stainless steel and titanium, different companies
- Structured undersurface → **limited contact** between plate and bone
- Even distribution of holes along plate
- → more periosteal blood supply preserved, smaller footprint

Feature	DCP	LC-DCP
Undersurface	Straight, no notches	Undercut / notched
Bone contact	Intimate, full	Limited
Stiffness	Holes less stiff than between holes	Even along length
On contouring	Bends at screw holes (kinking)	Lower kinking risk; more contourable
Fatigue at holes	Higher	Reduced
Screw inclination	Less	More, longitudinal and transverse
Periosteal blood supply	Compromised	Better preserved

Tubular Plates Ask AI

Tubular

- Famous **one-third tubular plate**
  - Lateral malleolus (classic)
  - Formerly medial condyle in distal humeral fractures
  - Bent as **spring plate** → posterior wall of acetabulum
  - Bent → buttressing effect on **quadrilateral plate**
  - Also distal ulna, olecranon
- One-third tubular → 3.5 system
- **Semi-tubular plate** → like one-third tubular, used with 4.5 screws
  - Historically forearm fractures (MCQs); lecturer never needed it
- Semi-tubular → limited stability, **collar holes**

Reconstruction Plates Ask AI

Recon

- Classic → pelvic and acetabular fractures
  - Deep notches between holes
  - Most important → **accurate contouring in any plane**
- Contoured with **bending pliers** (acetabular set)
- Other uses → distal humerus (newer plates exist), clavicle, olecranon
- Sizes → 3.5 and 4.5; straight and curved
- Specific plates for the **symphysis pubis** → Synthes and others

Spring Plate Ask AI

Spring

- **Anatomical spring plate** → Synthes and other companies
- Homemade spring plate from one-third tubular → posterior wall:
  - Cut with a cutter through last hole → bend cut ends → tines
  - Placed on very small **posterior wall** fragment that cannot hold screws
  - About 3 holes long → two screws above, none in fragment
  - Tines → prongs/hooks → capture posterior wall → reduction without screw
- Improvised ("homemade") vs anatomical spring plate

Plate	Sizes / screws	Classic uses
DCP broad 4.5	4.5	Femur; humerus; tibia
DCP narrow 4.5	4.5	Small humerus
Small DCP	3.5	Forearm
One-third tubular	3.5 system	Lateral malleolus, distal ulna, olecranon, medial distal humerus; spring plate
Semi-tubular	4.5 screws	Forearm (historical)
Reconstruction	3.5, 4.5; straight, curved	Pelvis, acetabulum, distal humerus, clavicle, olecranon

Locking Plates: LCP and LISS Ask AI

LCP Design Ask AI

LCP

- **LCP** (locking compression plate) ≠ LC-DCP (low-contact dynamic compression plate)
- Latest in evolution → described as an **internal fixator**
- Each hole = **combi hole** → locking and conventional screws
  - One part → conventional screw
  - Other part → threads for threaded locking screw heads
- "Internal fixator" → **extraperiosteal location** → no need for exact contouring
- Combines **DCP principle and locking head principle**
  - Flexibility of choice within a single implant
- Holes accept standard cortical screws or locking screws
- New AO plate standard → approved by **AO TK (Technical Commission)**, December 2000
- Locking compression plate → figure

Screw Types in the LCP Ask AI

- **Self-tapping screws** → metaphyseal areas
  - Require pre-drilling and length measurement → bicortical
- **Self-drilling, self-tapping screws** (incl. locking) → diaphysis, unicortical
- Last screw may be unicortical
  - **Periprosthetic fractures** → stem occupies canal → unicortical screws
  - **Bicortical fixation preferred** if achievable (periprosthetic webinar, YouTube channel)

Conventional versus Locked Fixation: The Hoop Ask AI

- Traditional plating → stability by **compression of the plate to the bone surface**
  - Bicortical screws → each engages two cortices
  - Two bicortical screws → **rectangular hoop**
- Locking screws → **angular stable fixation** within plate hole → similar hoop, unicortical
  - Two fixation levels → plate threads + near cortex
- Role of **unicortical fixation** → explained by angle-stable fixation

Feature	Traditional plating	Locked plating
Source of stability	Compression of plate to bone	Angular stability in plate hole
Screws	Bicortical	Unicortical possible
Two fixation levels	Near and far cortex	Plate threads and near cortex
Result	Rectangular hoop	Similar hoop

LCP as Internal Fixator Ask AI

- Bridges **multifragmentary or complex** metaphyseal or diaphyseal fracture
  - Not perfectly contoured → does not press on periosteum
- All locking screws → **internal fixator** ("internal external fixator")
- **Very versatile** → combi holes accept standard cortical and cancellous screws
  - Then a **traditional plate** → must conform to bone surface
  - Usually no longer used → locking screws preferred

Indications of Locking Plates Ask AI

- Not for all fractures → **special indications**
  - Expensive
  - Just a tool in the surgeon's hands → correct use → correct results
  - **Not forgiving or tolerant** like DCPs or other plates
- Classic indications:
  - **Osteoporotic bone** → impaired screw pull-out strength
    - Locking screws → reduce screw pull-out; usually elderly
  - **Non-unions** → local **disuse atrophy** / disuse osteoporosis
  - **Short segment of bone** → insufficient bicortical fixation, e.g. distal femoral condyles
    - Usually elderly → coexists with osteoporotic bone
  - **MIPO technique** (minimally invasive plate osteosynthesis)
    - MIO (minimally invasive osteosynthesis) → also includes intramedullary nail
  - **Periprosthetic fractures**

> **High Yield**
> - Locking plate indications:
> - Osteoporotic bone; non-unions; short bone segment
> - MIPO technique; periprosthetic fractures

Locking Plate Tips and Quotations Ask AI

- Not miraculous, not unbreakable → initial case broke ≈ 3–4 months post-op
- **Locking screws cannot be used for reduction** → fixation only
  - Not lag screws → may maintain compression achieved by other means
- "Reduce before locking" → after locking, no reduction possible
- "Lag before you lock" → after locking, no **lagging effect**
- **Osteoporotic bone** → usually relative stability (complex, multifragmentary)
- Lag screws in very osteoporotic bone → fail: screw strips or bone breaks
- "Don't mix principles" → relative stability → no lag screws
  - Some AO Trauma authorities → temporary lag screws, placed carefully, removed at end
  - Otherwise avoid lag screws → **relative stability concept**
- Initial case → **two lag screws** placed, yet plate broke

Must-Know

- "Reduce before locking"
- "Lag before you lock"
- "Don't mix principles" → no lag screws in relative-stability fixation
- Locking screws → never lag screws, never reduction tools

LISS Ask AI

LISS

- **Less Invasive Stabilization System (LISS)** → pre-shaped plates + self-drilling, self-tapping screws
  - Almost like locking compression plates
- Allows **MIPO technique** → small distal incision → plate slid into position
  - Minimally invasive ≠ small incision only → preserves soft tissue if done properly
  - **Image intensifier** → plate must lie on bone proximally
  - Fixed distally → plate may stand off proximally → **helicopter effect**
  - Proximal screws → outside bone or barely unicortical → less than optimum fixation
  - → **radiological checking** essential
- Intraoperative appearance → AO Surgery Reference

Pitfall

- MIPO plate fixed distally → stands off bone proximally → **helicopter effect**
- Proximal screws miss bone or barely unicortical → check radiologically

Reduction Aids Ask AI

- **AO femoral distractor** → distraction across fracture site
  - Classic indication → complex multifragmentary fractures
- Alternative → **Schanz screws** and external fixator
- Very useful reduction tool
- Plate applied with distractor kept → **maintain alignment**, adjust traction
  - Reduction not anatomical → restore length, axis, rotation
  - → check **AP and lateral views**

LISS Insertion Ask AI

- Through **jig** → screws inserted and locked via small incisions at each hole
  - Screws + plate → good **angular stability**
- Properly performed **MIO technique** → reduced surgical invasion
  - Preserves blood supply and soft tissue
- **LISS jig** → sleeve and drill bits; distal and proximal holes
- Proper implant position → confirm on **AP and lateral** views
- Distal femur → classic site
- Similar systems → other sites, incl. **proximal tibia**

Special Peri-articular Plates Ask AI

- Several **special peri-articular plates** → specific locations
  - Anatomically shaped → side-specific
  - Many with **combi holes** → versatile
- **Bending near a locking screw hole** → caution
  - May deform locking hole threads
  - → changes screw head–locking interface quality → **failure of the locking mechanism**
- Examples (AO book): DHS, distal femoral locking plate, distal lateral tibial plate
- Distal radius plates → incl. **fragment-specific plates**
- Plate history → beyond scope
- First case → examine all plates and screws; seniors show juniors

Plate Holes and Screw Insertion Ask AI

DCP Principle Ask AI

DCP hole

- Holes → among most important plate components
- **DCP principle** → screw holes **oval in shape** with a sloping part
- **Sloping part** → placed distant from the fracture
- DCP and LC-DCP → both have sloping form (LCP combi holes differ)
- Screw tightened → glides / slides → **compression**
- Sloping form → slightly different in DCP vs LC-DCP
- **Sloping shoulder** → part of an angled cylinder
  - Spherical screw head → moves down as screw tightened
- Spherical gliding principle → formerly **cylindrical gliding principle**
- **Eccentric insertion** → hole edge away from fracture (sloping part)
- Head glides down sloping shoulder → **moves the bone with it** relative to plate
- Bone moves → compression across fracture site
- On tightening → head forced down shoulder → bony fragment moves → fracture compressed

Neutral then Eccentric Screw Ask AI

- Plate applied → first **neutral screw** (central) on one side of fracture
  - Sleeve → green
- Formerly / some places → **neutral sleeve = green**; eccentric sleeve = gold / yellow
- Now also → **universal sleeve** (spring-loaded universal drill guide)
- Other side → eccentric screw → compression
- Yellow / gold sleeve → eccentric placement → glides down shoulder → bone moves → compression

Universal Drill Guide Ask AI

- **Spring-loaded** → serves as both neutral and eccentric guide
- Eccentric → no pressure; place at hole edge toward sloping shoulder; drill
- Neutral → press down spring mechanism → larger-diameter part centred in hole
- Comparison:

Universal guide state	Part in hole	Screw position	Equivalent standard sleeve
Not pushed down	Thin part protrudes → pushed to hole edge	Eccentric	Yellow / gold
Pushed down	Thicker, larger-diameter part	Neutral (centred)	Green

Drill Sizes for Lag Screws Ask AI

- Keep chart on mobile until memorised; lag technique → screws lecture
- Frequently asked in orals
  - **Gliding hole** = near cortex
  - Pilot hole = far cortex
- Drill sizes:

Screw	Gliding hole (near cortex)	Pilot hole (far cortex)
4.5 mm	4.5 mm	3.2 mm
3.5 mm	3.5 mm	2.5 mm
2.7 mm	2.7 mm	—

Exam Pearl

- 3.5 mm screw → pilot hole 2.5 mm
- Common error → 2.7 mm pilot (common hospital drill) → wrong
- 2.7 mm drill → gliding hole of the 2.7 mm screw

LCP Combi Holes Ask AI

LCP hole

- **Combi hole** → conventional screw in one part, with some angulation
- Common error → non-standard LCPs from some companies
  - Locking mechanism inappropriate → sleeve does not hold
  - **Discrepancy between threads and drill sleeve** → overuse or imperfect manufacture
- Drill sleeve → **absolutely perpendicular (100%)** to threaded hole
- Angled insertion → absolutely not
  - Optimal **angular stability** → locking screw at 90 degrees
  - Applies to standard LCP, not polyaxial or variable-angle systems
- 90 degrees → required for optimal angular stability

Variable-Angle LCP Ask AI

- Threaded holes with a range of angulation → **variable angle**
  - Classic → **distal radius plate**: radial styloid, distal radius at different angles
- Synthes **funnel-shaped** variable angle LCP drill sleeve → very versatile
  - Allows 15 degrees deviation from nominal trajectory of locking hole
- **Free-hand** variable angle LCP drill sleeve

Fixed-Angle Devices Ask AI

- Other plates → some angular stability
- 95-degree **angled plate** → one of the oldest
  - Used extensively by seniors → distal femoral fractures
  - Still a role → osteotomies, especially proximal femoral
  - **Fixed-angle device** → metaphyseal fractures, especially distal femur
  - Also → femoral osteotomies
- Formerly ordinary **condylar buttress plate** → distal femur
  - Now → threaded LCP holes → locking head screws → angular stability
  - Advantage → self-tapping, self-drilling screws → osteoporotic bone

Interim Recap Ask AI

- Covered so far:
  - Functions of the plate (enumeration)
  - AO principles of internal fixation
  - **Absolute and relative** stability and indications
  - Biomechanical background → Perren strain theory
  - Forms and types of plates, specialised plates
  - Plate holes → DCP vs LC-DCP; cylindrical (gliding cylindrical) principle
  - **LCP combi holes** → precautions, tips and tricks
- Next → the six plate functions
- Neutralization (protection), compression, tension band, buttress, bridging, reduction

Neutralization (Protection) Function Ask AI

Concept Ask AI

Neutral.

- "Neutralization" → **neutralizes the forces**
- "Protection" → protects against harmful rotational and axial forces on fracture
- First → **interfragmentary compression** with a lag screw
- One screw → cannot hold a diaphyseal fracture
  - Lag screw exposed to excessive rotational and axial forces
  - → **neutralization plate** (protection plate) needed
  - Neutralizes harmful rotational and axial forces → **bending, torsional loads**
- **Interfragmentary screw alone would not be able to withstand** these forces
- Plate **protects the interfragmentary compression lag screws** from large bending and torsional forces

Technique Ask AI

Neutral.

- Classic example → **ankle fractures**, fibular (lateral malleolar) fractures
  - First → anatomical reduction + interfragmentary compression via lag screw
  - Plate → not for compression; screws placed as **neutral screws**
  - Classic plate → one-third tubular
- Steps → reduce and hold → fix → apply plate
  - Sometimes → more than one lag screw
- Neutralization plate → **must be properly contoured** to bone shape
- **Reconstruction plates** as neutralization → very good contouring needed
  - Pelvic / acetabular sets → **aluminium templates** → model for bending
- Specialised **benders / pliers** for the acetabulum
- Eccentric screw → creates compression → avoid in neutralization
  - Use **drill sleeve** → neutral screw position
  - Eccentric attempt → plate may shift
- → drill guide recommended; plate contoured

Compression Function Ask AI

Concept Ask AI

Compress.

- Proper compression of bone fragments → **decreases the gap** bone must bridge
- → **absolute stability** → fracture components prevented from moving relative to each other
- Achieved through lag screws or plating techniques
- **Compression plating** → reduce and compress transverse or oblique fractures
- Lag screw → interfragmentary compression in intra-articular fractures, or with neutralization plate

Methods of Plate Compression Ask AI

- **Pre-bending of the plate** → very important concept
- External compression devices → **articulated tension (compression) device**
  - Compression (more common) and distraction
- **Dynamic compression** → oval holes with eccentric screw placement
  - Oval hole with sloping shoulder → DCP and LC-DCP

Application in Transverse Fractures Ask AI

- **Transverse fractures** → reduced and temporarily fixed with clamps if possible
  - Spanish or bone clamp
- **Forceps placed so they do not interfere with the planned plate position**
  - Conflict → re-reduction after achieving reduction
- Neutral screw on one side → **eccentric screw** on the other
- Sloping shoulder → always away from fracture
  - Eccentric screw tightened → bone moves → **fracture gap closes**

Pre-bending the Plate Ask AI

Pre-bend

- Junior belief → gap closes fully once screw tightened → not quite
- **Without pre-bending** → fracture closes immediately beneath plate (maximum compression)
  - Compression not evenly distributed → **opens on the other side**
  - "Neutral and eccentric screws placed well, fracture still open"
  - Check **intraoperatively** → evidence pre-bending not done
  - Exactly contoured plate → **gap in the opposite cortex**
  - Why (exam) → compression maximal beneath plate, not evenly distributed over fracture
- Solution → **overbending of the plate**
  - Centre of overbend → directly over fracture; away from fracture → fails
  - Centre **stands off about 1 to 2 mm** from reduced fracture surface; not more
- Tightening → compression on **opposite side** first, then beneath plate
  - Plate's tendency to return to original shape → resists opposite cortex opening
- Further tightening → overbent segment flattens against bone → gap beneath plate closes
- **As the load screw is tightened, the tension generated in the plate compresses the fracture evenly across the full diameter of the bone**

Exam Pearl

- Exactly contoured plate, no pre-bending → **gap in the opposite cortex**
- Reason → compression maximal beneath plate, not evenly distributed
- Fix → overbend centred over fracture line, stand-off 1 to 2 mm

Adding a Second Eccentric Screw Ask AI

- One eccentric screw, fracture not closed → first confirm pre-bending
- Insufficient compression after correct overbending → **some authorities** allow another eccentric screw
  - First eccentric screw → turned anti-clockwise → **loosened**
  - → tighten new one → retighten first → more compression
- Sequence → neutral → eccentric → (second eccentric after loosening) → all others neutral

Contouring Ask AI

- Contouring tools → most hospitals; very important for **DCP / large-profile plates**
- Neutralization or bridging plate → **bending pliers** or benders
  - Most LCPs → anatomically pre-shaped → usually unnecessary

Articulated Tension Device Ask AI

ATD

- Historical **external compression device** → asked in exams, fellowship examiners abroad
- Components:
  - Nut → tightened or loosened
  - Small **hook**
  - **Foot plate** with hole → screw inserted
- Technique → screws in other fragment → plate → hook in plate → foot plate screwed
  - Extra screw → must be removed later
  - **More exposure** → con
  - Nut tightened → hook pulls plate → **compression across the fracture site**
- Two modes:

Mode	Hook position	Nut	Result
Compression	In the plate	Clockwise	Compression across fracture site
Distraction	Against edge of plate, outside hole	Anti-clockwise	Distraction across fracture site

- Clockwise → bone moves → **compression mode**
- **Distraction** → used less
  - Some seniors ask about it; now rarely used → replaced by new plate designs
- After distraction → hook placed in plate → compression possible

Oblique Fractures: Creating an Axilla Ask AI

- Frequently asked in **FRCS** and fellowships
  - Device on side of acute or obtuse angle?
- Oblique fracture → device applied to **create an axilla** → fragments interlock
  - Same principle as dynamic compression plates
- **Acute angle** → axilla → fragments interlock, pressed together
- **Obtuse angle** → fracture may displace → deforming, not compressing, force *[C2 · transcript: "other side: acute angle or obtuse angle"]*
- Obtuse-angle side → marked X → avoid

Device side	Effect
Acute angle	Creates axilla → interlocking compression
Obtuse angle	Deforming force → displacement

- After compression → screws placed → device removed
  - Cons → screw removed leaving its hole; **approach extended** about 3–4 cm
- Replaced by **DCP plates** and compression plating techniques

Tension Band Function Ask AI

Principle Ask AI

T. band

- Fracture **loaded on either end** → fulcrum (bending point)
- Convex and concave sides:
  - **Concave side** = compression surface
  - **Convex side** = tensile (distraction) surface
  - Tensile forces → maximal on side opposite fulcrum
- **Tension band principle of plating** → plate opposite fulcrum, on convex side
  - **Applying the plate on the tension surface results in conversion of the tensile forces into compression forces**
- Plate as a band on tensile surface → compressive forces across fracture **under load**

Surface	Side	Forces
Concave	Fulcrum side	Compression
Convex	Opposite fulcrum	Tensile (distraction), maximal

The Femur Ask AI

- Femur → **eccentrically loaded bone**
  - Axial load → lateral (convex) cortex under tensile forces
  - Medial cortex → compressive forces
- Fracture → tensile forces open lateral side, medial side closes
  - Plate → **tension (convex) side** → distraction converted to compression *[C3 · transcript: "plate on compression convex side"]*
- Medial plate on femur → nothing counteracts convex-surface forces → **gapping**

Condition: Intact Opposite Cortex Ask AI

- Plate on tension side → can still fail
  - **Compression surface must be intact**
  - Comminution or wedge on compression side → failure under loading
  - **Implant failure may occur due to lack of buttress of the opposite cortex**
  - → **bone graft** to reconstitute opposite compression cortex
- Mid-shaft femoral fracture today → **intramedullary nail**, relative stability
  - Unless not indicated, e.g. children
- Compression side not intact → must be addressed (e.g. bone graft)

Pitfall

- Tension band plate on correct side still fails if **compression cortex** comminuted
- Cause → lack of buttress of the opposite cortex
- Remedy → bone graft to reconstitute the compression cortex

Definition and Ulnar Example Ask AI

- Definition → **a tension band** is a device placed on the tension side of an eccentrically loaded fracture surface, converting tensile force into compressive force
  - Always on tension side → prevents gapping
  - Medial (compression) side of femur → gapping
- Classic examples → **lateral femur**, posterior ulnar plate
- **Posterior ulnar plate** → comminuted ulnar / olecranon fractures
  - Plate and screws rather than cerclage wiring (formerly "tension band")
- Olecranon **deforming forces** → triceps and brachialis
  - Distract fracture complex → distractive forces on tensile side
- Plate on **dorsal aspect of the ulna** → distractive forces converted to compression
  - Analogous to plate on lateral aspect of femur

AO Update on Tension Band Wiring Ask AI

- Wiring → **previously thought of as a tension band** (AO)
  - Recent studies → may work simply via the **cerclage wire**
  - Compression occurs only in extension
- Former teaching → same principle in bony prominences
  - Olecranon, greater tuberosity, greater trochanter, lateral malleolus
- AO → **do not confuse the tension band plate principle with cerclage wire fixation**

Buttress and Anti-glide Function Ask AI

Concept Ask AI

Buttress

- **Buttress** → from French "to butt"
  - Also "to strike", "to shove", "to push", "to support" → pushing / pressing
- Diaphyseal fractures → **fractures under axial loads tend to shorten**
- Plate applied → **screws tightened distally**
  - Proximal part of plate, no screws → **buttressing effect** against other fragment
- **"The buttress plate acts like a thumb pressing the other fragment into a reduced position"**
- Finger pushes fragment back up the **inclined plane**
- Classic → **proximal tibial fractures**
  - Today → lag screws for intra-articular fracture + buttress plate

Functions and Indications Ask AI

- **Resists shear or bending forces** during axial loading
- Stabilises distal part or buttresses an **intra-articular fragment**
- **The plate must be contoured to fit the bone**
- Indications → metaphyseal fractures, AO **type B** (partial articular)
- Classic → **proximal tibial (tibial head)** and distal radial fractures
  - Volar distal radius → ordinary buttress plate, pre-locking era → sufficient
- Screws in buttressed part → not necessarily needed (image: **Rodrigo Pesántez**)
- Plus → absolute stability of intra-articular fracture via **interfragmentary compression**

Technique Ask AI

Buttress

- Older method, proximal tibia → **first screw closest to the fracture**
  - Tightened → upper plate buttresses upper fragment of proximal tibia
  - Then → **subarticular screws** (proximal tibial screws)
- Current → **lag screws** first, then buttress plate, medial or lateral aspect
- **Buttress plates are often used to supplement lag screw fixation of metaphyseal shear or split fractures**
  - **The screw may be inserted either through or outside the buttress plate**

Tibial Plateau Fractures Ask AI

- Medial tibial plateau or **bicondylar** fractures
  - Lag screw first
  - → buttress plate, contoured slightly **off the bone** → withstands shearing forces
  - Screw closest to fracture → plate presses **proximal fragment** → buttressing effect
  - Remaining screws → inserted normally
- Bicondylar tip → **reduce the medial tibial plateau first**
  - Usually less comminuted → one big chunk
  - Buttress plate → realign with shaft → then other side
  - **The plate resists displacement by axial loading**
- Second plate → lateral aspect

Anti-glide Plate Ask AI

Anti-glide

- **Anti-glide** → can be used interchangeably with buttress; famous AO / oral question
- Plate **posteriorly on the fibula** → large role in osteoporotic or rheumatoid patients
- Ankle fractures in rheumatoid, osteoporotic, osteopenic patients → very challenging
  - Now → **anatomical fibular (lateral malleolar) plates**
- Posterior plate → no screws in fragment; screws above
- Fragment too osteoporotic for screw → good **anti-glide effect**
  - Resists displacement and shear forces
- Difference → learned from **Ananda Nanu** (UK) → current BOA president
- Terminology:

Fracture	Plate called
Intra-articular	Buttress plate
Diaphyseal	Anti-glide plate

Exam Pearl

- Intra-articular fracture → **buttress plate**
- Diaphyseal fracture → anti-glide plate
- Full marks in oral exams

Bridging Function Ask AI

Bridge Plate Concept Ask AI

Bridge

- **Bridge plate** → multifragmentary or complex fractures with comminution
  - **Bridges the comminution with proximal and distal fixation** points (screws)
  - Fracture site left totally undisturbed
- Anatomical reduction attempt → **soft tissue** and blood supply disrupted
- Indication → **multifragmentary long bone fractures where intramedullary nailing or conventional plating is not suitable**
  - E.g. distal femur → though **distal femoral nails** now available
- **Bridging plate technique** → relative stability
  - Corrects length, alignment, rotation
  - Secondary healing → callus formation
- Whole fracture site **undisturbed** → preserves blood supply and soft tissue
- AO Surgery Reference → **biological osteosynthesis**
  - Restore bone length, axis, rotational alignment
  - Respect fragment biology → fracture zone not approached
- Indications → **diaphyseal fractures**, or intramedullary nails unusable
  - Classic → humerus, femur, phalanges
- Plate fixed above and below → healing
  - Formerly → slight curve in middle → **wave plate**

LCP as Bridge Plate Ask AI

- **LCP** with locking head screws → internal fixator → **no precise contouring required**
- LCP with conventional screws → **conventional bridge plate** → exact contouring required
- Locking head screw advantages:
  - **More stability** in osteoporotic bone → reduced screw pull-out risk
  - Reduced risk of **over-tightening** of screws
  - **Reduced fractures stay reduced**
- Locking screws ≠ lag screws → **"Lag before you lock"**
- **Unicortical screws** → end hole of plate, or periprosthetic fractures without bicortical option
- Unicortical locking → similar hoop to **rectangular hoop** of bicortical fixation
- **The plate does not need to be perfectly contoured** → internal fixator
  - **The plate is not pressed against the bone, and the periosteum is not compromised**
  - → smaller footprint, preserved blood supply

MIPO and Reduction Ask AI

Bridge

- Extent of approach → bridging plate via **MIPO technique**
  - E.g. jig of distal femoral locking plate
  - Humerus MIPO → **demanding technique**, attention to radial nerve
- Restore **length, axis, and rotation** first
  - External fixator, Schanz screws, AO femoral distractor
  - All tips and tricks → proper **functional reduction**
- **Key concept of bridge plating: the plate is fixed only above and below to the main bone fragments**

Perren Strain Theory Applied Ask AI

- Small gap without absolute stability → **high strain**
- Large gap without absolute stability → no high strain
- **Fracture zone left untouched** → maximizes blood supply
- Small gap, no absolute stability → eventual **plate failure**
- Large gap → greater working length, lower strain → failure much less likely

Working Length Ask AI

- Bridging → **long plates with a long working length**
- Long plate → **distribution of the bending forces** over longer segment
  - Lower stress per unit area
  - **Prevents high stress over the fracture site and reduces the risk of plate failure**
- Working length → should be large
- **Long plates allow a longer lever arm** → reduced screw pull-out, esp. osteoporotic bone
- Not every hole filled → several holes left open → **less stress at each hole** → breakage less likely
- Fixation → conventional screws in **neutral mode** or locking head screws (combi holes)
- **If you plan to lag, you have to lag first before you lock**
  - Lagging in bridging → absolutely not

High Yield

- Bridge plate → long plate, long **working length**
- Bending forces distributed → lower stress per unit area → less plate failure
- Longer lever arm → less screw pull-out
- Open holes → less stress at each hole

Plate Span Ratio Ask AI

- Final concepts → **plate span ratio** and plate screw density
- Very commonly asked → how long should the plate be?
- **Gautier and Sommer**, Injury, approximately 2003 → important paper to read
- **Plate length should be at least two and preferably three times the fracture length**
  - `10 cm` fracture → at least `20 cm`, preferably `30 cm`
  - Lecturer → always `30 cm` for `10 cm` fracture
  - In holes → at least three times as many holes as overlie fracture
  - Hole spacing differs between plates → calculate in centimetres, choose closest

Plate Screw Density Ask AI

- **The number of screws versus the number of screw holes should be less than 50%** (< 0.5)
  - Screw in every other hole → 50%
  - All distal holes filled → exception: distal segment too small to skip holes
  - Good **purchase** needed in both proximal and distal fragments
- Worked example:

Segment	Holes	Screws	Density
Proximal	6	3	50%
Middle (over fracture)	About 4	0	0%
Distal	4	3	75%
Whole plate	14	6	6/14 = 43% (0.43) *[C4 · transcript: "final screw density 0.43%"]*

- Distal 75% → accepted for anatomical reasons → no fewer than 3 screws possible
- Many seniors working in trauma → unaware of this
- Choose plate accordingly

High Yield

- Plate span ratio → plate ≥ 2×, preferably 3× fracture length (Gautier and Sommer)
- **Plate screw density** → screws / holes < 0.5 *[C5 · transcript: "screw density less than 0.5%"]*
- Example → 14 holes, 6 screws → 0.43

Reduction Function Ask AI

Reduction Plate Ask AI

Reduction

- Last function → **reduction**; added in the AO book
- Classic → 95-degree **condylar plate**
  - Formerly → proximal or distal femoral fractures; also osteotomies
  - Today → rarely used, except osteotomies
  - Plate as **reducing aid** → "reduction plate"

Provisional Reduction for Nailing Ask AI

- Source → article from "**10 Commandments of Open Fractures**" webinar
- **Intramedullary nail** without strong assistance, or proximal part keeps angulating
  - → proper reaming or reduction not achievable
  - Plate → one-third tubular or small DCP (holds itself somewhat), or 3.5 LC-DCP
  - → **temporary reduction tool** for provisional stabilization
  - Proximal and distal screws → **out of the trajectory of the after-coming nail**
- → reaming and nail passage in reduced position
  - Screws must not jeopardize nail trajectory or reaming
- In that study → plate may be removed afterwards
- Others → **does not compromise the results**
  - Sometimes left **definitively** alongside the nail

Summary of Plate Functions (AO Table) Ask AI

- AO book table → keep on mobile; review before each case

Function	Mechanism	Stability / healing	Classic example
Compression	Plate produces compression at fracture site	Absolute; primary healing, no callus	Simple transverse humeral fracture (lag screw not usable)
Protection (neutralization)	Neutralizes harmful bending and rotational forces → protects lag screw fixation	Absolute	Simple oblique radial fracture
Buttress / anti-glide	Resists axial load by applying force at potential deformity; counteracts shear	—	Lateral or medial tibial plateau fracture
Tension band	On convex (tensile) aspect → tensile forces converted to compression on concave opposite cortex	—	Lateral femoral plate; dorsal ulnar plate
Bridge	Spans comminution	Relative	Multifragmentary or complex; diaphyseal fractures
Reduction	Plate as reduction tool	—	Plate during intramedullary nailing (open fracture paper)

- Protection plate → lag screw first (insufficient alone), then plate
- Buttress → like placing our thumb to reduce fracture

Initial Case Revisited and Conclusion Ask AI

Failure Analysis Ask AI

- 74 years, rheumatoid, **very osteoporotic** → one of the most difficult cases
- Screws → reasonable; company may lack all sizes
  - Insist on **proper planning**; demand all required implants
  - Otherwise the patient pays
  - "What you tolerate is what will continue"
- Error 1 → **plate span**: very long fracture → plate ≥ three times its length needed
- Error 2 → **plate screw density**: every other screw should have been used
- Error 3 → **mixing principles**: lag screws placed
  - Lag screws in osteoporotic fractures → fail
  - Fracture cannot be anatomically reduced → may disrupt blood supply
- Weight bearing started → plate broke

Pitfall

- Why the initial plate failed:
- Plate too short → **plate span** less than three times fracture length
- Screw density too high → not every other hole
- Lag screws in osteoporotic bone → mixed principles

Revision Ask AI

- Alternative from start → **retrograde nail**, accepted; discussion limited to plating
- Patient counselled → plating chosen again (fracture to be opened)
  - Distal femoral nail → not a wrong choice
- **Medial comminution** → bone graft + double plating on medial femur
- Longer plate; **medial buttress plate** → only two screws needed
  - One screw placed close → could be omitted
  - No medial reduction attempted → only bone graft added
- Very osteoporotic at 74 years + rheumatoid → very difficult
- Lesson → that screw should have gone further up
  - **Medial approach** → reaches up to two-thirds of distal femur, sometimes limits
- Lateral view → osteoporotic bone → reduction in different planes difficult
  - Reduction tools / clamps → **crush and break the bone**
- Patient **lost to follow-up**
  - Seen up to 6 months → walking very well
  - Early healing; no compelling evidence of much bone formation

Conclusion Ask AI

- Lecture outcomes:
  - **Absolute and relative stability** → difference and indications
  - Simplified Perren strain theory
  - Plate functions → five, or six with reduction function
  - Forms, designs, evolution of plates
  - Difference between **DCP, LC-DCP, LCP, and LISS**
  - Tips and tricks
- Take-home → "The surgeon, not the designer of the plate, determines how a plate will function and how it will be applied."
- More important than fixation method → preserve bone's **soft tissue**
  - Bone–soft tissue attachment and blood supply
  - Do not treat X-rays → patient is not just a bone
  - Fracture = **a soft-tissue injury that happens to include a break of the bone**
  - "We are gardeners, not carpenters."

Questions and Answers Ask AI

Locking Screws and Type of Stability Ask AI

- Q (Dr Johnny): LCP for **dynamic compression** with locking screws in all holes?
  - Locking screws in all holes → no reduction (compression) achieved
  - Diaphyseal compression → locking plate not needed, except osteoporotic bone, periprosthetic fractures
  - Absolute stability → compression techniques discussed
  - Lag screw + **neutralization plate** feasible → plate may take locking screws (all act as neutralization)
  - Same for tension band principle
  - Sloping shoulder (dynamic compression) → neutral then eccentric screw → impossible with locking screw
- Q (Dr Tarek Hamza): LCP with correct plate span and screw density → absolute or relative?
  - **Absolute** → anatomical reduction, no micromotion across fracture site
  - Relative → multifragmentary part left undisturbed → bridging plate indication
  - LCP as **internal fixator** → relative stability
- **No such thing as "rigid stability"** → only absolute or relative
  - **Rigidity** → implant property: withstands deformation

Compression Technique and Angular Stability Ask AI

- Q (Dr Metwally Sayed): **eccentric screws** on both sides of transverse fracture?
  - Only on one side → classically neutral screw on one side eccentric on the other
  - More compression → additional eccentric screw on same side (debated by some)
  - First eccentric loosened → second tightened → first retightened → rest neutral
- Q: importance and indication of **angular stability**?
  - Full lecture on locking plates needed; indications = locking plate indications
  - Osteoporotic fractures, periprosthetic fractures, MIPO technique
  - Angular stability → reduces screw pull-out (increases pull-out strength)
- Q (Dr Ahmed Mahmoud): **over-compression** through all DCP holes → jeopardizes or delays union?
  - Proper compression → one neutral + one eccentric screw on the other side + overbending
  - Several eccentric screws → probably not needed
- Q (Dr Ahmed Abu Salem): first screw nearest to or away from fracture?
  - Dynamic compression → neutral screw **adjacent to the fracture**; other likewise nearest to fracture
- Q (Dr Amr Abul-Naga): does **cerclage wiring** transform deforming force into compression?
  - Yes → performed by the cerclage itself, not exactly the tension band principle

Working Length and Plate Choice Ask AI

- Q: anatomical reduction with absolute stability → longer plate with free holes (**working length**)?
  - No → anatomical reduction → no callus needed → **primary bone healing**
  - No need for working length → screws adjacent to fracture (transverse or oblique)
  - Multifragmentary fractures → working length matters
  - Improper fixation leaving a **gap** → relative stability with very short working length
  - → non-union or delayed union, or eventual plate breakage or failure
- Q (Dr Mostafa Shawky): AO images → **simple shaft radius fracture**, 7-hole plate, exactly 3 holes each side
  - That fixation = **absolute stability**; relative only if multifragmentary or comminuted
- **Neutralization** → oblique fracture with lag screw, insufficient alone with one or two screws
  - Examples → lateral malleolar fracture; humerus with short oblique or spiral fracture
  - Neutralization plate → **still absolute stability**, not relative
- Q (Dr Amr Ibn Nagar): **LC-DCP** replaces DCP in all situations to reduce periosteal injury?
  - New plates → mostly LC or LCP, depending on companies
  - Definite advantage → **minimized footprint** of LC-DCP vs DCP
  - Egypt → not available in all hospitals or for all patients (cost)
  - Proper indication + **proper soft-tissue handling** → DCP results good
  - Failures, infection, non-healing → improper soft-tissue handling and technique

Reduction, Forearm and Plate Design Ask AI

- Q (Dr Omar El-Sisi): **positional screw** through bridging plate to reduce big fragments?
  - Yes, depending on **bone quality** → as temporary reducing aid, then removed
  - Especially osteoporotic → **mixing the principles** → not ideal
  - Cerclage → possible; preferably with **wire passer** (ordinary passage strips more)
  - Usually unneeded → plate itself via MIPO; AO femoral distractor; Schanz screws
  - Plate itself (**LISS**) → reduce fracture → fix in reduced position
  - Several ways → avoid positional screw in a butterfly fragment
- Q (Dr Amr Abul-Naga): two eccentric screws instead of **neutral and eccentric**?
  - Classic → probably no more needed; may cause **distracting force** on first side
  - Convention → neutral on one side, eccentric on the other
- Q (Dr Mohamed El-Ashry): **forearm fractures** → treated exactly like intra-articular fractures
  - **Anatomical reduction** → restores supination and pronation
  - Multifragmentary → absolute stability not achievable
  - Radius and ulna reducible anatomically → absolute stability a must
- **We usually treat both-bones forearm fractures as intra-articular fractures**
- Q (Dr Ahmed Abu El-Fadl): pre-bending → not for reduction here
  - Traditionally → plate curved around the area → **wave plate**
  - So-called **biological plating** → ordinary plates passing outside malaligned, protruding fracture
  - Needed in every case → no
- Q (Dr Ayman El-Sheikh): **plate thickness and profile** → factor in resisting rotational forces?
  - Probably → more rigid implant withstands more force
  - Different metallurgy → locally manufactured implants in Egypt
  - Probably a role against failure; no **biomechanical study** cited
- Remaining questions → send after video upload to YouTube

Final Questions and Closing Ask AI

- Q (Dr Hisham Abu Omar): DCP → neutral screw in proximal or distal fragment?
  - Either main fragment → no problem
  - Only consideration → **tension device**: acute-angle side, not obtuse
- Q (Dr Mazen): eccentric screw farther from fracture instead of **pre-bending**?
  - Not sure → lecturer follows classic **overbending of the plate** (AO technique)
  - Farther placement → probably still **fracture gapping**; no definitive answer
- Q: long fractures → single plate or **overlapping double plates**?
  - Lecturer → **nails**, not plates
  - Overlapping double plate only if no nailing facilities, nailing not feasible, or plate too short
  - → should be **predicted preoperatively**
  - Outside AO principles → overlapping double plates in **specific indications**
- Moderator thanks → on behalf of **Viatris** company
- Lecturer thanks Mina, Viatris, **technical assistance**
- Future lectures → **screws and different principles**